A3 pelvis alignment

Current v3 marker shell, pelvis-aligned OptiTrack asset and optional measured calibration.

Current hardware

Use **mocap_sticker_shell_v3**, with **17 stations**: seven front markers and two rear rows of five. Station identities remain **S04-S08 and S11-S22**. The separate paddle mount is **right_wrist_paddle_flange**.

The sticker optical-centre table uses metres in **pelvis_link**: **X forward, Y left, Z up**. Its points include the accepted **0.20 mm** sticker-plus-adhesive thickness. The printable shell and flange STLs use millimetres at 100% scale.

Choose the intended localization workflow

Mode	Input and fixed correction
motive_live Default	Uses the selected body's current NatNet ModelDef and imported pelvis pivot/axes. Assumes identity local correction. No calibration JSON or fixed file/layout pin is required.
motive_asset Optional asset check	Checks the supplied 17-marker asset against its pinned definition. Follow the component guide for the configured asset and identity checks.
calibration Optional measured correction	Explicitly enables the measured marker-to-CAD workflow. Saves an installation-specific receipt and uses its accepted fixed local correction.

In all modes, the world pose changes as the robot moves. Identity describes only the intended **fixed local UCB_P1-to-pelvis_link correction**. World/table registration is a separate operation.

Hardware index and current procedures
agibot/README.md

ASSET PREPARATION

Keep the pelvis pivot

Use the current shell, asset and coordinate convention together.

1. Install the matching layout
Use the v3 printable halves and all 17 sticker stations. Front: S11, S12, S13, S19-S22. Rear: upper S04-S08 and middle S14-S18. A 12 mm ball needs its own measured sphere-centre offset; the sticker table is not a ball-centre table.

2. Use the current native asset
Import **A3_v3_17_stickers.motive**, internally named **UCB_P1**. Use **UCB_P2** for the other table side. Confirm the installed layout, marker count, streaming identity and preserved pelvis pivot. The companion CSV is a coordinate reference, not the native import file.

3. Keep the stream axes consistent
Set Motive Streaming **Up Axis = Z Up**. Retain **topics.rigid_body_markers.modeldef_y_up_to_z_up: true**. Native asset offsets use Motive's storage basis; the bridge conversion produces ROS X forward, Y left, Z up.

4. Preserve the aligned pivot and orientation
Do not use Reset Location / Reset Orientation after importing the aligned asset. Do not add an old centroid offset or a second pose rotation. The intended fixed correction is translation [0, 0, 0] m and quaternion [0, 0, 0, 1] in xyzw order.

5. Refresh and verify the live definition
Restart the NatNet adapter after asset changes. Check the selected body and live tracking using the current operator procedure. The five restored rear points have nominal CAD coordinates but no observations in the original source export; those coordinates alone do not prove visibility.

Shell files, station evidence, asset and frame conventions
agibot/mocap_sticker_shell_v3/README.md

Canonical sticker optical centres
agibot/mocap_sticker_shell_v3/documents/marker_transforms_pelvis_link_stickers.csv

Wrist paddle mount and printing notes
agibot/right_wrist_paddle_flange/README.md

OPTIONAL MEASURED MODE

Measure the fixed correction

Run this workflow only when selecting localization_source:=calibration and start_calibration:=true.

Build and source the adapter and HOPE workspace, enable the rigid-body marker topic with **publish_ucb_markers:=true**, and reach the stationary **PD_STAND** setup state through the selected operator procedure. Then run from the external computer's repository root:

```
source hope_ws/install/setup.bash
ros2 run hope_bringup p1_marker_cad_calibrator \
  --topic /optitrack/rigid_body_markers \
  --asset-name UCB_P1 \
  --marker-layout stickers_v3 \
  --table-side P1 \
  --minimum-frames 200 \
  --capture-duration 4 \
  --stationary-prepare \
  --attest-installed-layout \
  --output calibration/ucb_robot_to_pelvis.json
```

What the capture establishes

The solver matches live body-local ModelDef points to the CAD sticker centres and checks physical labeled-marker samples. All 17 stations must be defined; each needs enough valid observed samples during capture, without requiring simultaneous visibility. Marker stream order alone does not identify physical stations.

The solver checks its fitted correction; it does not force identity. An accepted fit replaces the laptop-local receipt atomically. A failed capture leaves the previous receipt untouched. Earlier ten-marker, 24-station and 12-point receipts do not qualify the current sticker layout.

How the result is used

The relay composes the accepted fixed correction with the live world-to-UCB body pose. It publishes **/a3/base_pose_flat** and diagnostic **/a3/mocap/pelvis_pose**. The robot receives the live topic. A saved stationary world-pelvis snapshot is an audit record, not the world pose of a moving robot.

Complete capture procedure and acceptance gates
agibot/README.md#v3-sticker-p1-to-pelvis-calibration

Receipt lifecycle, identity defaults and migration rules
calibration/README.md

Keep the two methods distinct

The older pose-pair method remains an independent audit, separate from the current sticker capture workflow.

Legacy independent pose-pair audit

p1_pelvis_calibrator compares a live mocap body pose with an independently produced, full-6DOF pelvis pose at matching acquisition times. Both inputs must share a known reference frame and clock epoch. The independent reference must not be derived from the mocap body or the transform being measured.

The A3 pelvis IMU supplies orientation and inertial data, not absolute arena translation. An independent external metrology system, state estimator or correctly aligned simulation must supply that full pose. This method requires translation and rotation excitation; its checks differ from the stationary sticker capture on the preceding page.

The independent audit does not verify the installed per-sticker layout or generate an approved production sticker receipt. Follow the current measured-calibration procedure when that receipt is required. The default motive_live workflow remains independent of a saved calibration JSON.

Frames and correction ownership

pelvis_link is the A3 physical root frame. **/a3/calibration/pelvis_pose** is a temporary independent-reference PoseStamped topic, not a TF frame. Apply a fixed local correction once through the selected runtime path; do not duplicate a Motive pivot correction in an additional ROS transform.

Authoritative technical references

OptiTrack setup and optional measured/independent calibration
docs/OPTITRACK.md

Shared mocap contract
mocap/README.md

Coordinate-frame contract
docs/interfaces/frames.md

Current operator and console workflows
foxglove/README.md

This guide reflects the checked-in 17-station hardware and localization modes. Use the component READMEs and interface contracts for executable commands, current parameter definitions and detailed acceptance criteria.